1. C
2. D
3. A
4. A
5. D
6. B
7. D
8. B
9. A
10. C
11. B
12. B
13. A
14. C
15. B
16. B
17. C
18. 4213
19. D
20. C
21. 2431
22. 2431
23. A
24. A

1. A
2. 4
3. C
4. 4
5. C
6. B
7. D
8. A
9. B
10. B
11. A
12. 16
13. D
14. B
15. 8
16. 340
17. A
18. 366
19. C
20. 3

1. A
2. C
3. C
4. 25
5. 161
6. D
7. B
8. 2
9. A
10. 364
11. A
12. B
13. B
14. A
15. 172
16. C
17. D
18. 4
19. C
20. C
21. 20
22. B

Solutions for questions 1 to 4:

Number of words and Explanatory notes for RC:

Number of words: 538

1. Option A: Scientists do not work with unambiguous facts, they have to place facts in context, to interpret them within a particular framework or theory. In cosmology or atomic physics, this process is relatively uncontroversial. The first part of choice A is true. But when we are dealing with the question of human nature / origins, the creation of the framework within which one places the facts can be contentious, and open to political, philosophical and cultural influences. So, it's not that the facts are ambiguous, it's that we look at them through coloured lenses. So, the second part of choice A is distorted.

Option B: In the former, scientists find it difficult to know which facts are important and which irrelevant, unless they already had a framework or theory into which they could fit the facts. So, choice B should have been read as: In the former, facts are twisted to suit theories.

Option C: Scientists do not work with unambiguous facts, they have to place facts in context, to interpret them within a particular framework or theory. Scientists find it difficult to know which facts are important and which irrelevant, unless they already have a framework or theory into which they could fit the facts. In cosmology or atomic physics, this process is rather uncontroversial. .... When we are dealing with the question of human nature (origins), the creation of the framework within which one places the facts can be contentious, and open to political, philosophical and cultural influences.

Hence choice C is correct and is the answer.

Option D: In the former, one theorizes before one has data because they already have a theory or framework into which they could fit the facts. Therefore, this is not true. Besides, the second part of choice D has not been mentioned in the passage. Hence, this is not the answer.

Choice (C)

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2. Option A: A narrative passage tells a story, usually from one person's viewpoint. A narrative passage is a sequence of events and follows a storytelling format. A narrative passage neither presents any analysis nor evokes any emotions. This passage is not narrative. Hence choice A is not the answer.
   Option B: Analysis involves examining aspects of a situation in its plusses and minuses, and making an evaluation at the end of it. In this passage, the author mentions his viewpoint in the first para: When we are dealing with human nature or human origin, the framework is open to political, philosophical and cultural influences. He backs it up with the views of Jared Diamond and Robert Foley. While it does seem that the author is analysing the arguments that have been presented, he's actually laying the ground for an argument of his own (at the end). The author's style is not analytical but argumentative. Hence choice B is not the answer.
   Option C: A descriptive passage makes a discussion vivid with detail. Here the author is not using a descriptive tone. Hence choice C is incorrect.
   Option D: The passage is argumentative. An argumentative passage presents arguments and counterarguments. The author in the passage delivers an argument and he tries to convince us of his argument. Refer to para 3: In adopting the second argument … what it means to be human onto the data. Refer to para 4: Today's insistence that humans are nothing …. disillusionment with such opinion. The purpose of the last para is to offer the author's argument: We have to understand ...... Putting the two together will tell us about humanness and the present state of humanity. Overall the style of the passage is argumentative. Hence choice D is correct.
   Choice (D)

3. Refer to para 2.
   Option A: Choice A is Diamond's opinion. Diamond's argument is that 'what we really are', the essence of humanity, has little to do with conventional indices of humanness: language, culture, technology, and so on. Rather it is expressed principally through our ***animal heritage*** (inherent traits correspond to animal heritage). Hence, this is the answer.
   Option B: According to Diamond, if the conventional indices like language, culture, technology etc. are taken away, then what is left is humanity, the essence of which is expressed principally through our animal heritage. Therefore, we have our humanity even if these things are taken away. Hence, it can be inferred that Diamond believed that these things are not the distinguishing features of humanity. This makes choice B wrong.
   Option C: The genetic proximity of Man and ape is without question. But choice C is not the author's opinion.
   Option D: According to Diamond, if we "take away our clothes, possessions and language", we look like apes. However, the vice versa cannot be implied from this. Hence, Choice D is not the answer.
   Choice (A)

4. Option A: The key issue for many Darwinists is the similarity of humans and apes. Darwinian explanations also draw on philosophical and cultural assumptions about what constitutes humanity, how humans relate to the non-human animal world and so on… ***The roots of our behaviour must lie in our animal, and in particular ape, ancestry***. Hence choice A is the correct answer.
   Option B: Choice B is not the story that Darwinists would tell. Choice B does not discuss the view of Darwinists at all. It only explains that the original Victorian image of man is reversed today: Today there is a 'loss of confidence in the extent to which humans could be said to be on a pedestal above the swamp of animal brutishness.' The almost boundless capacity of humans to do damage to each other have, in the twentieth century, rather dented human self-esteem. ..... Apes have become more angelic during the course of the twentieth century, the angels, or at least their human representatives, more apish. ..... Originally humans were thought to be the advanced form of life (the angels), and other animals the more primitive. Hence choice B is incorrect.
   Option C: The animal within us is our noble side, and humanity or civilization the darker side – a complete reversal of the original Victorian image. Choice C again does not give us the viewpoint of Darwinists.
   Option D: Today's insistence that humans are nothing more than another kind of ape is the consequence of a century's worth of disillusionment with such optimism. But choice D is far-fetched and is not the answer as it does not pertain to the view of Darwinists as discussed in the passage. Had the statement been 'The story of the divergence of the human and ape evolutionary tree lines' it would have been worthy of consideration.
   Choice (A)

Solutions for questions 5 to 8:

Number of words and Explanatory notes for RC:

Number of words: 544

5. "A set of straight and gleaming teeth makes for a beautiful smile. But how many people who have undergone a little dental maintenance know that they may have inside their mouths some of the first products of a new industrial revolution?" The author then goes on to introduce the concept and technology of 3D printing to us.
   Option A: The condition "unless they have a beautiful smile" in choice A makes it incorrect.
   Option B: The passage does not say that 3D printing is complicated as a concept and people do not and cannot understand it. Hence choice B is incorrect.

Option C: The word “only” in choice C makes it a narrow choice. We can rule it out as this has not been implied in the passage.

Option D: But how many people who have undergone a little dental maintenance know ..... products of a new industrial revolution? Tens of millions of dental crowns, bridges and orthodontic braces have now been produced with the help of additive manufacturing or 3D printing. From this, it can be inferred that people are not aware of 3D printing and its applications. Hence choice D is the correct answer. Choice (D)

6. Option A: While it has been mentioned in para 1 that 3D printing technology is used to produce dental crowns, bridges and orthodontic braces, and further in para 3 that things are done differently at an industrial unit in Miskin, near Cardiff, we cannot infer that people at Miskin are more prone to dental problems. Choice A is incorrect.

   Option B: The plant is equipped with three of the firm's 3D printers; more will be added soon. Once complete, the parts are shipped to dental laboratories all over Europe where craftsmen add a layer of porcelain. Some researchers are now working on 3D printing the porcelain, too. From this, we can infer that more 3D printers will be required in future and that the demand is only increasing. Hence, choice B is the answer.

   Option C: We cannot say that employees at the British Engineering Company are not efficient in managing 3D printers. The printers run unattended and make each individual tooth to a design that is unique to every patient. The 3D printing process is successfully carried out here. Hence choice C is incorrect.

   Option D: “Existing 3D printers becoming defunct or running out of order” is a pessimistic view that has not been mentioned in the passage

   Choice (B)

7. Option A: The application of 3D printing in the medical field has been discussed in para 4. “Hundreds of thousands of people have been fitted with 3D-printed orthopaedic implants, from hip-replacement joints to titanium jawbones, as well as various prosthetics. An untold number have benefited from more accurate surgery carried out using 3D-printed surgical guides; around 100,000 knee replacements are now performed this way every year.” This does not mean that 3D printing carries out surgeries, but that it provides resources used in the carrying out of surgeries. So option A is out of scope.

   Option B: We cannot say that 3D printing was already on par with and has now replaced all conventional methods or processes. Choice B is far-fetched.

   Hence choice C is incorrect.

   Option D: It has been mentioned in para 3 that the 3D printers make each individual tooth to a design that is unique to every patient. Para 5 states that the health-care industry has so swiftly adopted additive manufacturing should be no surprise. People come in all shapes and sizes, so the ability of a 3D printer to offer customised production is a boon. In para 4, we are told that a large number of people have benefited from more accurate surgery carried out using 3D-printed surgical guides. Hence choice D is correct. Choice (D)

8. (a) In para 4 it has been mentioned that more than 60m custom-shaped ***hearing-aid shells and ear-moulds*** have been made with 3D printers since 2000. Hence, (a) is an example of 3D-printed industrial products already in use. (a) is correct and is not the answer.

   (b) Titanium jawbones is an example of the 3D-printed orthopaedic implant mentioned in para 4. Hence, (b) is correct and is not the answer.

   (c) Hip-replacement joints are mentioned as examples of the 3D-printed orthopaedic implants mentioned in para 4. As for knee-replacement, we are not told about the replacement components, but that 3D-printed surgical guides are used for more accurate surgeries. Refer to the last sentence of para 4: An untold number have benefited from more accurate surgery carried out using 3D-printed surgical guides; around 100,000 knee replacements are now performed this way every year. It cannot be inferred that these are knee replacement joints. So, (c) is partially incorrect.

   (d) Forget the idea of hobbyists printing off small plastic trinkets at home. Plastic trinkets is not an example of an industrial 3D-printed product. Hence, (d) is incorrect.

   (e) Alcoa, a leading producer of aluminium, recently said ***it would supply*** Airbus with 3D-printed titanium fuselage parts, 3D-printed fuel nozzles for jet engines and 3D-printed pylons used to attach engines to wings. This is an indication of what intends to be done, and is not enough to tell us that they're currently in use. So, (e) is incorrect.

   (f) ***In the future***, large 3D printers will be used not only to print smartphones and portable consumer electronics. There isn't enough to indicate that smartphones are a current application. So, (f) is also incorrect.

   Choice (B)

Solutions for questions 9 to 12:

Number of words and Explanatory notes for RC:

Option C: While the passage implies that 3D printing is cheap (And software is faster and **cheaper** to change **than tools used in a traditional factory**, which is designed to churn out identical products), it does not discuss the 3D printing business from the investors' perspective.

Number of words: 523

9. Option A: If we go back to where the author mentions the single lucky incident, we can read that the author says, this incident makes it easier for us to ***underestimate*** the multitude of ways luck had influenced the eventual outcome. In other words, the author says that there are many ways luck played a part in Google's fate, but we ignore it, and this one lucky incident makes it that much easier to ignore other parts where luck played a role. Option A is the answer.

Option B: This was true, but this is stating the obvious. This option doesn't tell us about the author's purpose in explaining to us this incident, and the author's takeaway. Option B is not the answer.

Option C: This is a sweeping generalisation. While it could be said that the author believed luck favored Google, there is no grand generalisation that luck mattered more than anything else in Google's success. Option C is not the answer.

Option D: This can be inferred from the para, true. However, the author didn't mention this example as a proof that luck had helped Google through every decision. The author mentioned this example to show how luck's role is often underestimated in Google's success. The incident shows Google was lucky, early on. The author mentions the incident to show how luck was there everywhere, and yet, those who believe the story hold on to a story like this and underestimate the luck elsewhere. Option D is not the answer.

Choice (A)

10. Option A: This is the WYSIATI rule that the author mentions in the first line of the para. However, it doesn't explain what the paradox is, that comes later in the para. Option A is not the answer.

Option B: While the author does talk about our ability to ignore our ignorance, the option doesn't indicate why the author uses the term 'paradoxically' or what the paradox here is. Option B is not the answer.

(not just events that contributed to Google's rise). Hence, Option A is not the answer.

Option B: The understanding according to the author of someone who reads the story is illusory because '*the ultimate test of an explanation is whether it would have made the event predictable in advance.*' Since, no narrative for Google's story could have possibly predicted the success in advance, this test cannot be passed. Hence, Option B is the answer.

Option C: The author doesn't say that the story is selective and leaves out the lucky bits. The author is simply explaining that any story created to explain Google's success will not be able to predict Google's success (without the hindsight bias). In other words, any story or narrative would explain Google's success post-facto, but the same narrative may not have been able to predict Google's success, because it is impossible for a narrative to involve all possible events that contributed to Google's success.

Option D: The author doesn't create two sets of events, one part where luck played a role, and another where luck didn't. The illusory nature of understanding is not because we know one part of the story and don't know about the other. It is illusory because the story cannot predict Google's success, and that's because the story won't include all the details that would have led to other outcomes. Hence, Option D is not the answer.

Choice (B)

12. Option A: The last line of the para indicates that the world makes sense because we are ignorant of our ignorance. Ignorance of our own ignorance is the foundation of our conviction it says. Hence, Option A can be understood from the para. It is not the answer.

Option C: This is the paradox, an obvious contradiction between two ideas that seem to occur together. The paradox here is that when we have less information, it is supposed to puzzle us more. However, when we have less information, we end up creating a more coherent (clear, and less puzzling) story or narrative. Hence, Option C is the answer.
Option D: This option states the obvious – fewer pieces, more coherence. However, it doesn't explain what the paradox (or contradiction was). This is because of a slight inaccuracy in this option. Fewer pieces in a puzzle obviously make the puzzle easier. The paradox is that when the puzzle is bigger, but ***only a few pieces are available***, it shouldn't offer more coherence (because there are many missing pieces). Hence, Option D is close but not the answer.

Choice (C)

11. Option A: The understanding is not illusory because we don't know about every event. It is illusory because it doesn't make the event predictable, and it doesn't make the event predictable because no story can include all the events that could have changed the outcome
Option B: This cannot be inferred from the para, because the author doesn't really question ***the reliability of our evidence***. The author merely questions the ***limited amount of information*** (quantity) that we have, which may not be sufficient to explain an event. Hence, Option B is the answer.
Option C: From the first line of the para, it is clear that we are ignorant of our own ignorance (reiterated in the last line again). So, it is indeed possible that we don't understand our lack of information while explaining something. Option C is not the answer.
Option D: The author does explain that we try to create a story believing that whatever information we have is all the information that exists. That means we can create coherent stories (even if they are not true) without all the information, and hence, it can be said that the coherence of the story we create is not necessarily correlated to the information (we may create a perfectly coherent, albeit wrong, story, with very little information). Option D is not the answer.

Choice (B)

Solutions for questions 13 to 16:

Number of words and Explanatory notes for RC:

Number of words: 509

13. Option A: The idea is that the consequences of policy should not privilege any conception of the good or conceptions of ultimate values. This conception, which has been attributed to some liberals, is incoherent. Through examples, the author reiterates that a policy of indifference would necessarily privilege one side or the other, as there is no third way. Hence a policy outcome cannot clearly be neutral as it is bound to favour one or the other side of the divide in any contested area of public policy. Choice A is the answer.
Option B: Choice B may be the fallout of what is mentioned in para 2 but it is not a statement about the error in ideology.
Option C: The author brings in impracticability but not rationality into this discussion. Hence, Option C is not the answer.
Option D: This is a distortion of the author's statement about a liberal policy of toleration (not of neutrality): A liberal policy of toleration has the effect of privileging the views of tolerant liberals over those of non-tolerant non-liberals. Liberal egalitarians are more concerned with neutrality in justification. Choice (A)

16. Rawls mentions that if people have full knowledge about themselves and their position in society, then they would make agreements that are biased in terms of self-preference and partiality. They also tend to enter agreements thinking of maximizing their own advantage/profit. He feels that this individual decision making should be filtered out and fairness can only be achieved this way.
Option A mentions that self-aware people tend to bargain in a way so as to attain maximum profit. This is something Rawls explicitly stated and can be gathered from the line "However, if people if people are allowed to enter the agreement with full knowledge of their own position in society, they will have a tendency to bargain to maximize their own advantage". So, it would be something Rawls would agree with and so Option A is not the answer.
Option B states that partiality can be avoided only when people are self-aware in terms of their position in the society. This is the exact opposite of Rawls' idea and can be understood from the line "The initial agreement takes place behind the veil of ignorance which filters out from individual decision-making the biases of self-preference and partiality". This is something Rawls would not agree with and hence, Option B is the answer.
Option C says that a fair basic structure can be achieved when people who are not self-aware enter into agreements only to improve their quality

14. Option A & D: Choices A and D cannot be inferred from the passage. These do not refer to the expression mentioned in the passage.
Option B: Choice B is incorrect. The expression "neutral about ultimate ends" in the passage means the ends delivering no particular privilege to anyone.
Option C: Choice C is the most correct. "Because the liberal state does not take a substantive view …… neutral about ultimate ends." It does not presuppose any basic values or conceptions of the good. Choice (C)

15. Statement I: If people are allowed to enter the agreement with full knowledge of their own position in society, then they will have a tendency to bargain to maximize their own advantage. The author also refers to the bias of 'self-preference'. Statement I can be inferred.
Statement II: Liberalism does take a view on the core ethical significance of individuals (para 3). Liberalism must give priority to equality of concern and the idea of the separatedness of persons – that individuals have ultimate moral significance and cannot be sacrificed for the good of others (para 4). Statement II can be inferred.
Statement III: Rawls' basic intuition is that fairness or impartiality can be achieved by combining ignorance ***with self-interest***. If we do not know who we are, but we are motivated to improve our condition in life, we will choose Rawls' principles of justice. Statement III is correct.
Choice (B)

of life. This can also be understood from the line "If we don't know who we are, but we are motivated to improve our condition in life, we will choose Rawls' principles of justice". This would be something Rawls would agree with and so, Option C is not the answer.
Option D mentions that if individual decision making is filtered out, it will lead to fair agreements. This is also something that can be gleaned from the line "The initial agreement…and partiality". Rawls would agree with this and so Option D is not the answer. Choice (B)

Solution for question 17:

17. Option A: This option picks the two examples mentioned in the para (which should generally be avoided in a summary, generally, unless they are essential to the theory being discussed). It doesn't give us the essence of the para which is the distinction between the two kinds of authorities. Option A is not the answer.
Option B: The para discusses what epistemic and administrative authorities are. It doesn't try to discuss the overlaps or who is in one category and not in the other. Option B is not the answer.
Option C: This represents the two types of authority discussed in the para, along with the specialty of each of them. Administrative authority is powerful because it is empowered by the law. Epistemic authority is wielded by someone who has expertise or knowledge and hence, we would do well to follow what they are trying to say. Option C is the answer.

Option D: While the greater part of the option depicts the content in the para, the part that brings in comparison - 'we are more likely to comply' - is misplaced. Such a comparison hasn't been made in the para, that we comply to one more than the other. Option D is not the answer. Choice (C)

Solution for question 18:

18. Sentence 1 starts with a positive connector 'Similarly'. It needs a precedent.
Sentence 2 is an independent sentence that mentions a theory that emotional contagion is for real.

Solution for question 20:

20. Option A: Physical and social environment doesn't reduce mortality and morbidity rates. These are factors that augment the rates and understanding these factors helps us counter that. Option A is not the answer.
Option B: While this option is true, it only represents a small part of the para – that physical and cultural conditions play a big role in dealing with the mortality and morbidity rates. It doesn't mention the genetic factors. It doesn't mention how all these factors can be understood

Sentence 3 refers to a preceding sentence with 'this network effect'.

Sentence 4 is an independent sentence that explains how emotional states can be transferred from one person to another.

It can be understood that 42 is a logical block, as 4 is upstream to 2, introducing emotional contagion (sentence 2 gives additional information about emotional contagion). 2 and 1 make up a mandatory block because of the similarity in the content, emotional contagion and moods transferred through social networks. That leaves us with Sentence 3. It refers to a network which has been discussed only in 1. So, 4213 is the order. Ans: (4213)

Solution for question 19:

19. The statements immediately preceding and following blank (1) talk about children's opportunities to play while the highlighted sentence talks about adult-directed activities. It therefore breaks the flow of the two statements. Hence, the highlighted sentence does not fit in blank (1).

The sentence immediately preceding blank (2) talks about children's free play in the first half of the 20th century. The sentence immediately following blank (2) talks about children having a good deal of free time in 1900 (first half of the 20th century). Therefore, these two statements are connected and do not require another sentence in between them. Hence, the highlighted sentence does not fit in blank (2).

The sentence immediately preceding blank (3) talks about children having a good deal of free time in 1900. The sentence immediately following blank (3) talks about how adults began chipping away at the freedom starting around 1960. Therefore, these two statements are connected and the highlighted sentence would break the flow. Hence, the highlighted sentence does not fit in blank (3).

The sentence immediately preceding blank (4) talks about how adults started chipping away at the freedom children had. The highlighted sentence elaborates on this by explaining how the freedom was chipped away by replacing children's games with adult-directed activities. "The changes" in the statement following blank (4) alludes to the replacement of children's games with adult directed activities. Hence, the highlighted sentence fits in blank (4). Choice (D)

to lower the same rates. Option B is not the answer.

Option C: It is true that knowledge of the three parameters: physical and cultural conditions, genetic similarities (what is general for the species) and genetic variations (what varies genetically), which augment mortality and morbidity rates, can be used to fight the same. So, this option represents the essence of the para, which is to talk about the factors that affect human well-being (measured by the rates of mortality and morbidity) and how to reduce those rates based on such an understanding of the factors. Option C is the answer.

Option D: While this statement is not an inaccurate description of part of the idea in the para, it leaves out the main idea, which is how to counter the mortality and morbidity rates based on an understanding of the parameters which augment/increase these rates. Option D is not the answer. Choice (C)

Solutions for questions 21 and 22:

21. Sentence 1 carries forward another idea using 'in this respect' stating that Audioslave has defied convention. So, in a previous sentence we have to discuss convention or what's the norm.

Sentence 2 introduces Audioslave and is an independent sentence.

Sentence 3 is an independent sentence that states a norm using the expression 'all too often'.

Sentence 4 is an independent sentence that talks about Chris Cornell stepping into the shoes of La Rocha.

It is easy to pair up 3 and 1, 3 stating the norm, and 1 continuing with how Audioslave is defying convention. 4 has to follow 2 because La Rocha was introduced in 2. 4 also continues the anticipation established in 2. So, 2431 is the order. Ans: (2431)

22. Sentence 1 starts with a positive connector 'too' and talks about the idea of a financial crisis that ends American dominance.

Sentence 2 is an independent sentence that asserts the importance of the evolution of credit and debt.

Sentence 3 gives examples of the importance of corporate finance – Dutch, British empires, as well as the US.

Sentence 4 gives another example of the importance of finance.

Since 2 is the biggest of all the ideas, the broad definition, it has to come first. While both 3 and 4 are lateral arguments (examples), it must be noted that 1 which definitely ends the para (because it talks about the decline of the US empire that too because of finance) needs its preceding sentence to be 3. That is because of the US link – the second half of 3 and Sentence 1 talking about the rise and probable fall of the US. So, 2431 is the order. Ans: (2431)

Solution for question 23:

23. The statement immediately preceding blank (1) talks about how gaslighting is just starting to be studied using systematic social scientific data. The highlighted sentence states that the data unfolds as part of an unequal social context, thus elaborating on what was observed when gaslighting was studied. Hence, the two statements are connected. The statement immediately following blank (1) talks about social vulnerabilities and stereotypes, thus building on the unequal social context mentioned in the highlighted sentence. Hence, the highlighted sentence fits in blank (1).

The statements immediately preceding and following blank (2) talk about gaslighting in the social context. The highlighted sentence mentions how gaslighting is perceived as a problem between two people. This does not fit between the said statements refer to the social context. Hence, the highlighted sentence in blank (2) would break the flow. Hence, the highlighted sentence does not fit in blank (2).

The highlighted sentence talks about how gaslighting also unfolds in the social context. The statement immediately following blank (3) talks about how it is also used to describe structural racism, sexism, homophobia and ableism. The highlighted sentence requires some elaboration that is provided in neither the statement immediately preceding blank (3) nor the statement immediately following blank (3). Hence, the highlighted sentence does not fit in blank (3).

The statements immediately preceding and following blank (4) talk about gaslighting in specific contexts like race. The highlighted sentence is more general in nature. Inserting the highlighted sentence in blank (4) would thus break the flow. Hence, the highlighted sentence does not fit in blank (4). Choice (A)

Solution for question 24:

24. Option A: This is the apt essence of the para which talks about two ideas: that people should be able to live in conditions that encourage them to pursue worthwhile lives, and secondly, democracy is most reliable when it comes to creating those conditions. The rest of the para revolves around these two ideas. Option A is the answer.

Option B: The aim of a state (and probably politics) is to provide a just social and political order and not to propagate 'morality' through the social and political order. It is a misrepresentation of the content. Hence, Option B can be eliminated.

Option C: The political arrangement, e.g. democracy is what gives us the state/government. The para doesn't say anywhere that the state creates the political arrangement. Hence, Option C is incorrect.

Option D: This option tells us what a democratic government should or shouldn't do. However, the para is not about a democratic government. It is about any government. In fact, in democracy, according to the para, the most reliable conditions reside to create conditions. Option D is not the answer. Choice (A)

Difficulty level wise summary - Section I

Level of Difficulty	Questions
Very Easy	–
Easy	–
Medium	1, 3, 5, 6, 10, 17, 18, 19, 23
Difficult	4, 7, 8, 9, 11, 12, 20, 21, 22, 24
Very Difficult	2, 13, 14, 15, 16

Solutions for questions 1 to 5:

At the beginning of the day, all the ferries were in Jarpur. From the given information, all the ferries were in the same city after Trip 10. For all the ferries to be in Jarpur at the end of the day, each ferry must have completed an even number of trips. For all the ferries to be in Harpur at the end of the day, each ferry must have completed an odd number of trips. Since the total number of trips is 10, and the sum of three odd numbers cannot be ten, each ferry must have completed an even number of trips. All the ferries must have been in Jarpur at the end of the day.

Hence, at the end of Trip 10, all the ferries must have been in Jarpur.

Before Trip 10, Ferry 3 must have been in a different bank to Ferry 1 and Ferry 2 (since only Ferry 3 can transport eight passengers). Hence, Ferry 3 must have been in Harpur before Trip 10.

Let the columns in the following table denote the two banks (J denotes Jarpur and H denotes Harpur). Before Trip 10, this must be their positions:

J	H
F1	F3
F2	

In Trip 9, 7 persons were transported. To know the locations of the ferries before this trip, we can move one ferry from one bank to another and see what the result will be.

If we move F1 from Jarpur to Harpur, then if seven persons came to either bank, it will not result in the above arrangement. If we move F2 from Jarpur to Harpur, then if seven persons came to the bank in which F2 and F3 are present, then we get the arrangement depicted above. If we move F3 from Harpur to Jarpur, then if seven persons came to the bank in which all three ferries are there, they would have taken F2, which is will not result in the arrangement.

Hence, before Trip 9, the only possible arrangement is this:

J	H
F1	F3
	F2

In Trip 8, 9 persons travelled. Hence, before Trip 8, there is only one possible arrangement:

J	H
F1	F2
F3	

In Trip 7, 5 persons travelled. If we move F1 from Jarpur to Harpur, then the five persons could have travelled using F1 in Trip 7 and we get the above arrangement. For the other two ferries, we get different arrangements. Hence, the only possible arrangement for before Trip 7 is this:

J	H
F3	F1
	F2

In Trip 6, 7 persons travelled from one bank to another. The only possibility is if F2 is in Jarpur before Trip 6. We get the following arrangement for before Trip 6:

J	H
F2	F1
F3	

In Trip 5, 6 persons travelled. We get two cases for before this trip. F2 can be in Harpur or F3 can be in Harpur.

J	H
F3	F1
	F2

J	H
F2	F1
	F3

In Trip 4, 6 persons travelled. In the first case above, the only possibility is if F2 is in Jarpur. In the second case above, the only possibility is if F2 is in Harpur. Corresponding to the two cases above, we get the following arrangements for before Trip 4:

J	H
F2	F1
F3	

J	H
	F1
	F2
	F3

In Trip 3, 8 persons travelled. This implies that F3 is the only ferry that could have travelled. We get the following arrangements for before this trip:

J	H
F2	F1
	F3

J	H
F3	F1
	F2

Since 7 persons travelled in Trip 2, we get the following possibilities for before Trip 2:

J	H
	F1
	F2
	F3

J	H
F3	F1
F2	

Before Trip 1, all the ferries must be on the same side. This is possible only in the second case. From this, we can eliminate the first case in each of the above trips. The following table provides the details of the trips that happened during the day:

Trip	**Number of Persons**	**Ferry**	**Origin**
Trip 1	5	Ferry 1	Jarpur
Trip 2	7	Ferry 2	Jarpur
Trip 3	8	Ferry 3	Jarpur
Trip 4	6	Ferry 2	Harpur
Trip 5	6	Ferry 3	Harpur
Trip 6	7	Ferry 2	Jarpur
Trip 7	5	Ferry 1	Harpur
Trip 8	9	Ferry 3	Jarpur
Trip 9	7	Ferry 2	Harpur
Trip 10	8	Ferry 3	Harpur

1. Trip 7 was made by Ferry 1. Choice (A)

2. The maximum number of trips made by any ferry is 4 (Ferry 2 as well as Ferry 3). Ans: (4)

3. Trip 8 was from Jarpur to Harpur. Choice (C)

4. The number of passengers who travelled from Jarpur to Harpur = 5 + 7 + 8 + 7 + 9 = 36
The number of passengers who travelled from Harpur to Jarpur = 6 + 6 + 5 + 7 + 8 = 32
Required difference = 4. Ans: (4)

5. After Trip 6, two ferries were in Harpur. Choice (C)

Solutions for questions 6 to 10:

From the given information, the five agents are the ones who did not use the name James, Jeeves, Julian, Joss and Joss when introducing themselves. Let these five agents be labelled Agent 1 to Agent 5 (for easy reference).

From (i), Agent 1 introduced himself as Jack to Agent 3. Hence, Agent 3 could not have introduced himself as Jack to Agent 1. From (iv), Agent 3 introduced himself as Jack and Jeeves to Agent 4 and Agent 5 (it does not matter who is which as Agent 4 and Agent 5 are indistinguishable at this point). We can assume that Agent 3 introduced himself as Jack to Agent 4 and as Jeeves to Agent 5. Hence, Agent 4 cannot introduce himself as Jack to Agent 3 and Agent 5 cannot introduce himself as Jeeves to Agent 3. Also, from (iv), Agent 4 introduced himself as Julian to Agent 1. Hence, Agent 1 cannot introduce himself as Julian to Agent 4. Since Agent 3 has introduced himself as Jack and Jeeves, he must have introduced himself as Joss and James to Agent 1 and Agent 2 in any order.

The agent in point (iii) introduced himself as Joss and as Jeeves. Hence, this cannot be Agent 2 or Agent 4 or Agent 5 (since they did not use these names for introduction). This also cannot be Agent 3 since he did not introduce himself as Joss to either Agent 4 or Agent 5. Hence, this must be Agent 1. Agent 1 must have introduced himself as Jeeves to someone and this agent must have introduced himself as Joss. Agent 1 could not have introduced himself as Jeeves to Agent 4 or Agent 5 (since they cannot introduce themselves as Joss). Hence, Agent 1 must have introduced himself as Jeeves to Agent 2. Agent 1 must have introduced himself as Jeeves to Agent 2 and Agent 2 must have introduced himself as Joss to Agent 1.

Agent 1 must have introduced himself as Julian and Joss to Agents 4 and 5 in any order. But he cannot introduce himself as Julian to Agent 4. Hence, Agent 1 must have introduced himself as Joss to Agent 4 and as Julian to Agent 5. Agent 5 cannot introduce himself as Julian to Agent 1.

Three agents must have introduced themselves using the same name to one of the agents. If this is Agent 1, then Agent 5 must have introduced himself as Joss to Agent 1. Since this is not possible, this cannot be Agent 1.

If this is Agent 2, three agents must have introduced themselves as Jeeves or James. Since Jeeves is not possible, three agents must have introduced themselves as James. If Agents 3, 4 and 5 introduced themselves as James to Agent 2, Agent 2 could not have introduced himself as James to any agent. Hence, this is not possible.

If this is Agent 3, three agents could have introduced themselves as Jack or James (Julian is not possible as Agent 4 already introduced himself as Julian to Agent 1). Since Jack is not possible, three agents must have introduced themselves as James. Agents 2, 4 and 5 introduced themselves as James. This is possible.

If this is Agent 4, three others do not have any names that they can use for introducing themselves.

If this is Agent 5, then three others must have used the name Julian. However, this is not possible as Agent 4 has used the name Julian for introducing himself to Agent 1.

Hence, the only possibility is that Agents 2, 4 and 5 introduced themselves as James to Agent 3. Since Agent 2 introduced himself as James to Agent 3, Agent 3 cannot introduce himself as James to Agent 2. The only possibility is Joss. Agent 3 must have introduced himself as James to Agent 1.

The remaining possibilities for the names are present in the table below:

	Agent 1	**Agent 2**	**Agent 3**	**Agent 4**	**Agent 5**
Agent 1	X	Jeeves	Jack	Joss	Julian
Agent 2	Joss	X	James	Jack/Julian	Julian/Jack
Agent 3	James	Joss	X	Jack	Jeeves
Agent 4	Julian	Jeeves/Jack	James	X	Jack/Jeeves
Agent 5	Jeeves/Jack		James		X

Agent 1 cannot be the agent to whom two agents introduced using the same name. Also, Agent 5 is definitely one of the agents to whom two agents introduced using the same name (for any of the cases).

Also, Agent 2 could not have introduced himself as Jack to Agent 4 since Agent 3 introduced himself as Jack to Agent 4. This is because if Agent 2 introduced himself as Jack to Agent 4, then Agent 2 and Agent 3 both would have introduced themselves as Jack to Agent 4, which violates (ii) (as the name Jack was not used by the two agents who used the same name to introduce themselves).

Hence, Agent 2 must have introduced himself as Julian to Agent 4. Agent 2 must have introduced himself as Jack to Agent 5.

Again, for Agent 5, since two agents cannot introduce themselves as Jack to Agent 5, Agent 4 must have introduced himself as Jeeves to Agent 5. From this, we can say that Agent 5 cannot introduce himself as Jeeves to Agent 4. Since Agent 5 already used the name James and does not use the name Joss, Agent 5 must have introduced himself as either Jack or Julian. However, if Agent 5 introduced himself as Jack to Agent 4, then both Agent 3 and Agent 5 would have used the name Jack to introduce themselves to Agent 4, which violates (ii). Hence, Agent 5 must have used the name Julian to introduce himself to Agent 4.

Also, Agent 4 must have introduced himself as Jack to Agent 2 (since he introduced himself as Jeeves to Agent 5). Since Agent 4 introduced himself as Jack to Agent 2, Agent 5 cannot introduce himself as Jack to Agent 2 (because of (ii)). Hence, Agent 5 must have introduced himself as Jeeves to Agent 2 and as Jack to Agent 1.

The following table provides the list of names that each agent used for introducing themselves:

	Agent 1	**Agent 2**	**Agent 3**	**Agent 4**	**Agent 5**
Agent 1	X	Jeeves	Jack	Joss	Julian
Agent 2	Joss	X	James	Julian	Jack
Agent 3	James	Joss	X	Jack	Jeeves
Agent 4	Julian	Jack	James	X	Jeeves
Agent 5	Jack	Jeeves	James	Julian	X

6. Agent 3 introduced himself to Agent 1 as James. Choice (B)

7. The two agents who did not use the name Joss are Agent 4 and Agent 5. They used the names Julian and Jeeves for introducing themselves to each other. Choice (D)

8. Three agents used the name James to introduce themselves to another agent. Choice (A)

9. Agent 3 is named Jeff. Jeff introduced himself as Jack and Jeeves to Agent 4 and Agent 5. Choice (B)

10. Agent 1 is Jerome. Jerome introduced himself as Jack to Agent 3. Choice (B)

Solutions for questions 11 to 15:

11. Let the first person choose any chair. The second person can sit two places to the right or two places to the left of this person. Assume that the second person sat two places to the right of the first person. The third person can sit three places to the right or three places to the left of the second person. To minimize the number of chairs, we can consider the third person to be sitting three places to the left of the second person. That is, the third person will be sitting to the immediate left of the first person. Similarly, we can accommodate all the ten persons in eleven chairs as shown below:

9	7	5	3	1		2	4	6	8	10

Hence, a minimum of 1 empty chair will be present for seating 10 persons. Choice (A)

12. Let the first person choose any chair at random. Assume that the second person sits two places to the right of the first person.
Consider that the third person sits to the immediate left of the first person. The fourth person sits to the immediate right of the second person. From the given condition, the fifth person has to sit five places to the right of the fourth person.
The sixth person cannot sit six places to the left of the fifth person (as this place is occupied by the second person). Hence, the sixth person must sit six places to the right of the fifth person. To minimize the number of chairs, the seventh person can sit the immediate left of the fifth person and the eighth person can sit to the immediate right of the sixth person and so on. This is presented below and in this case, a total of 18 chairs are required.

3	1		2	4		11	9	7	5						6	8	10

However, there is another case possible. Let the first, second and third persons sit in the same way as in the previous case but in the other direction. The fourth person can sit four places to the right of the third person. The fifth person must sit five places to the right of the fourth person (as they are on the same side). The sixth person can sit to the immediate left of the fourth person. The seventh person can sit to the immediate right of the fifth person and so on. This case is presented below:

2		1	3	10	8	6	4					5	7	9	11

In this case, only 16 chairs are required.
Hence, a minimum of 16 chairs are required for 11 persons. Ans: (16)

13. Option A: It is possible that only two chairs are unoccupied. One way is shown below.

1	4	2			3	5

Option B: It is possible that five chairs are unoccupied. One way is shown below.

5				1	4	2			3

Option C: It is possible that eight chairs are unoccupied in the following way:

5					4				3	1		2

Hence, all of the given options are possible. Choice (D)

14. Since the seventh person sat in the 7th chair, the sixth person must be sitting in the 14th chair. The eighth person must be sitting in the 15th chair. Similarly, the fourth and second persons will be sitting in the 13th and 12th chair respectively.
The first, third, fifth, seventh and ninth persons will be sitting in the 10th, 9th, 8th, 7th and 6th chairs respectively.
The total number of chairs required will be 15.
Choice (B)

15. Given that the third person sat in the chair numbered 3. We need to minimize the number of the chair in which the ninth person sits. The second person must sit in the 6th chair, the 4th person must be in the 7th chair. From here, we can take the 5th, 6th and 7th person away from the 4th person and then move back along the line for the 8th and 9th persons. The 5th person will sit in the 12th chair, the 6th person will sit in the 18th chair, the 7th person will sit in the 25th chair. The 8th person can sit in the 17th chair and the 9th person can sit in the 8th chair. This is the closest that the 9th person can sit to the first chair.
Hence, the minimum number of the chair in which the ninth person sits is 8. Ans: (8)

Solutions for questions 16 to 20:

16. The total number of SUVs sold during the given period = 100 + 120 + 160 + 180 + 120 + 80 + 100 + 40 + 80 + 100 = 1080.
Total number of Sedans sold during the given period = 160 + 100 + 180 + 160 + 120 + 140 + 180 + 120 + 160 + 100 = 1420.
Required difference = 340. Ans: (340)

17. Given that 40% of the SUVs were sold at a discount. The number of SUVs sold at a discount in the ten months were 40, 48, 64, 72, 48, 32, 40, 16, 32 and 40, respectively.
The number of Buses sold in the ten months were 60, 100, 40, 80, 80, 40, 100, 120, 20 and 40, respectively.
Hence, the number of SUVs sold at a discount was greater than the number of Buses sold in two months, i.e., March and September.
Choice (A)

18. The total number of vehicles sold at a discount in January = 40% × 100 + 30% × 160 + 25% × 180 + 10% × 60 = 40 + 48 + 45 + 6 = 139.
Total number of vehicles sold at a discount in February = 40% × 120 + 30% × 100 + 25% × 80 + 10% × 100 = 48 + 30 + 20 + 10 = 108.
Total number of vehicles sold at a discount in July = 40% × 100 + 30% × 180 + 25% × 60 + 10% × 100 = 40 + 54 + 15 + 10 = 119.
Total number of vehicles sold at a discount in the three months = 366. Ans: (366)

19. Total number of SUVs sold = 1080 of which 432 would have been sold at a discount.
Total number of Sedans sold = 1420 of which 426 would have been sold at a discount.
Total number of Trucks sold = 960 of which 240 would have been sold at a discount.
Total number of Buses sold = 680 of which 68 would have been sold at a discount.
The percentage of vehicles sold at a discount $= \frac{432 + 426 + 240 + 68}{1080 + 1420 + 960 + 680} = 28.16\%$
Choice (C)

20. In all the months in which more than 400 vehicles were sold, for the number of vehicles sold at a discount to be less than 100, the percentage of vehicles sold at a discount must be at most 25%.
Hence, in these months, there should be a high proportion of Trucks and Buses sold (since only these vehicles can bring the percentage down to 25%).
However, in all these months (January, March, April, May and July), the number of SUVs and Sedans is close to 50%. Hence, we can eliminate these months.
In February, the total number of vehicles sold at a discount = 108 (from the previous solution).
In June, the total number of vehicles sold at a discount = 32 + 42 + 10 + 4 = 88.
In August, the total number of vehicles sold at a discount = 16 + 36 + 15 + 12 = 79.
In September, the total number of vehicles sold at

2. It is given that the triangle has two equal sides of 10 cm each. Let the third side (i.e., the base) of the triangle be 2b.
Area of the triangle $= \frac{1}{2} \times base \times height = \frac{1}{2} \times 2b \times \sqrt{10^2 - b^2} = b \times \sqrt{100 - b^2}$
It is given that the area of the triangle is less than 48 sq. cm.
Hence, $b \times \sqrt{100 - b^2} < 48 \Rightarrow b^2(100 - b^2) < 48^2$
$\Rightarrow b^2(100 - b^2) < 2304$ ---------- (1)
The options provide four values for the length of the third side (i.e., 2b). We can substitute different values of b provided in the options to check which one does not satisfy the above inequality.
The inequality is satisfied for all the values except 14 (i.e., for b = 7).

Alternative Solution 1:

a discount = 32 + 48 + 30 + 2 = 112.
In October, the total number of vehicles sold at a discount = 40 + 30 + 25 + 4 = 99.
Hence, less than 100 vehicles were sold at a discount in three months, June, August and October. Ans: (3)

Difficulty level wise summary - Section II	
Level of Difficulty	Questions
Very Easy	–
Easy	11, 16, 17, 18, 19, 20
Medium	1, 2, 3, 4, 5
Difficult	12, 13
Very Difficult	6, 7, 8, 9, 10, 14, 15

Solutions for questions 1 to 22:

1. Given, $-4 \le x \le 3$; $5 \le y \le 11$.
By trial and error, we can find the minimum values of the expression as follows:

(1) Similarly, the minimum value of $x^3y - xy^3$ i.e., $xy[x^2 - y^2]$
$= 3 \times 11[9 - 121] = -3696$

(2) Given $x^3y - xy^2 = xy[x^2 - y]$
The minimum value $= -4 \times 11[(-4)^2 - 11]$
$= -44 \times 5 = -220$

(3) Given $x^2y + xy^2 = xy[x + y]$
The minimum value $= -4 \times 11[-4 + 11]$
$= -44 \times 7 = -308$

(4) Given $xy^2 + x^3y = xy[x^2 + y]$
The minimum value $= -4 \times 11[(-4)^2 + 11]$
$= -44 \times [16 + 11] = -44 \times 27 = -1188$

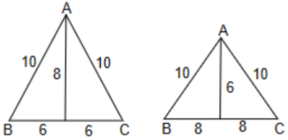
∴ The minimum value of the expression in choice (A) is the least.

Choice (A)

By substituting $b^2 = x$ in the inequality (1) in the above solution, we get
$x(100 - x) < 48^2 \Rightarrow x^2 - 100x + 48^2 > 0$
$\Rightarrow (x - 64)(x - 36) > 0$
$\Rightarrow x < 36$ or $x > 64 \Rightarrow b < 6$ or $b > 8$.
Hence, the third side (i.e., 2b) is less than 12 or greater than 16.

Alternative Solution 2:

Consider the following two possibilities.



In either case the area of the triangle ABC is 48 cm².
Also ∠BAC in the two figures are supplements. Let us denote the vertex angle in the figure on the left as α. ∴ The one on the right is π – α.
The area of ΔABC is $\frac{1}{2}$ (AB) (AC) sinθ, where θ can vary.
If θ < α, area of ΔABC < 48
If θ = α, area of ΔABC = 48
If α < θ < π – α area of ΔABC > 48
If θ = π – α, Area of ΔABC = 48
If θ > π – α, Area of ΔABC < 48
As the area of the triangle is given to be less than 48 cm²,
θ < α or θ > π – α
⇒ BC < 12 or BC > 16. BC cannot be 14.

Choice (C)

3. The value of f(x) exists for all values except
$12 - \lfloor x \rfloor = 0$
$\Rightarrow \lfloor x \rfloor = 12$
i.e., for $x \in [12, 13)$
∴ Range = (–∞, 12) ∪ [13, ∞) Choice (C)

Thus, the speed of the stream was 25% of the speed at which he was rowing the boat.

Ans: (25)

5. The arithmetic progressions given are
$A_1 \rightarrow$ 7, 13, 19, 25, 31, 37, 43, ……

4. This question is best solved using the concept of relative speeds. The speed of the boat in still water is its speed relative to the water. This relative speed remains the same, whether the boat is travelling upstream or downstream. Also, when the hat floats along with the water, its speed with respect to the water will be zero. Hence, the situation can be thought of as the boat moving away from the stationary hat at a certain speed (i.e., the speed of the boat relative to the water) for some time and then returning to the hat at the same speed, and hence taking the same amount of time. Hence, the boat have turned back when the hat had drifted a distance of exactly $\frac{2x}{2} = x$ (with respect to the ground). Therefore the hat drifted a distance x downstream in the same time that the man rowed the boat a distance of $3x$ upstream (both with respect to the ground). Hence $\frac{u-v}{v} = \frac{3}{1}$, where u and v are the speeds of the boat with respect to water and the speed of the water respectively. Hence $\frac{v}{u} = \frac{1}{4}$ i.e., v is 25% of u.

Alternative solution:

Let the distance travelled upstream after dropping the hat be m and the distance the hat travels downstream till he retrieves it be n (i.e., $m = 3x$ and $n = 2x$).

Considering his rowing speed and the speed of the stream to be u and v respectively, we get

$\frac{m}{u-v} + \frac{m+n}{u+v} = \frac{n}{v}$

$\Rightarrow (mu + mv + mu - mv + mu - nv)v = n(u^2 - v^2)$

$\Rightarrow uv\,(2m + n) = nu^2$

$\because u \neq 0$

$\therefore nu - v(2m + n) = 0$

$\Rightarrow nu = v(2m + n)$

$\Rightarrow \frac{u}{v} - \frac{2m+n}{n}$

It is given that $n = \frac{2}{3}m \Rightarrow m = \frac{3}{2}n$

$\therefore \frac{u}{v} = \frac{2\left(\frac{3}{2}n\right)n}{n}$

$\Rightarrow \frac{u}{v} = 4$ i.e., $v = \frac{1}{4}u$

$\therefore \frac{u}{v} \times 100\% = \frac{1}{4} \times 100\% = 25\%$

$A_2 \rightarrow 5, 13, 21, 29, 37, 45, \ldots\ldots$

We can see that the first term common between the two series is 13. Since the common differences of A_1 and A_2 are 6 and 8 respectively, any two consecutive terms common between A_1 and A_2 differ by LCM (6, 8) i.e., 24. So the series of common terms also form an arithmetic progression. The series is 13, 37, 61, ….

Given that t_n of A_1 + t_n of A_2 = 9000

$7 + 6(n - 1) + 5 + 8(n - 1) = 9000$

$14n - 2 = 9000$

$n = \frac{9002}{14} = 643$

t_{643} of $A_1 = 7 + (643 - 1)6 = 3859$

t_{643} of $A_2 = 5 + (643 - 1)8 = 5141$

So all the terms common to A_1 and A_2 will be less than or equal to 3859. The number of terms common to A_1 and A_2 is same as the number of terms in the series 13, 37, 61, …. less than or equal to 3859

i.e., $\left[\frac{3859-13}{24}\right] + 1 = \left[\frac{3846}{24}\right] + 1 = 161$

Ans: (161)

6.

X	Y	Z	U	V
5	18			
	2	25		
		60	0.8	
			51	68

Equating the ratios, we have

x	y	z	u	v
5	18	225	3	4

$5x \equiv 3u \Rightarrow 17x \equiv 3 \times \frac{17}{5} = 10.2u$

$18y \equiv 225z$

$\Rightarrow 90y \equiv 225z \times 5 = 1125z$

$4v \equiv 5x \Rightarrow 36v \equiv 45x$

$18y \equiv 4v \Rightarrow v$ is greater than y.

$\therefore$ $28y$ cannot be equal to $126v$. Choice (D)

7. The probability of either a 2 or a 3 turning up for a die

$= \frac{2}{6} = \frac{1}{3}$

The probability that neither a 2 nor a 3 turning up

$= 1 - \frac{1}{3} = \frac{2}{3}$

The probability that neither a 2 nor a 3 turning up on any of the three dice = $\frac{2}{3} \times \frac{2}{3} \times \frac{2}{3} = \frac{8}{27}$

The probability that either a 2 or a 3 turning up on at least one of the dice = $1 - \frac{8}{27} = \frac{19}{27}$.

Choice (B)

8. The number is divisible by 100. As it has only two prime factors, its prime factors must be 2 and 5. Let the number be in the form $2^a\,5^b$ where a and b are natural numbers.

The number would be $100(a + 1)(b + 1)$

$2^a\,5^b = 2^2\,5^2\,(a + 1)(b + 1)$

It follows from the equation above that $a \geq 2$ and $b \geq 2$.

If $a = 2$ or $b = 2$, the R.H.S. of the equation above would be divisible by 3. while the LHS would not be.

$\therefore a > 2$ and $b > 2$.

If $a = b = 3$,

L.H.S. = 1000

R.H.S. = 1600 ≠ L.H.S.

$\therefore a = b = 3$ is not possible.

If $a = 4$, $b = 3$,

L.H.S. = R.H.S. = 2000.

For any further increase in any of a or b, the L.H.S. increases faster than the R.H.S. Hence $a = 4$ and $b = 3$ is the only solution possible. And the number = 2000, i.e., the sum of its digits = 2.

Ans: (2)

9. The second piece of work takes 6 man days x 3 x 3 = 54 women days.

Now, since 10 + 9 + 8 + 7 + 6 + 5 + 4 + 3 + 2 = 54, the work takes 9 days to complete.

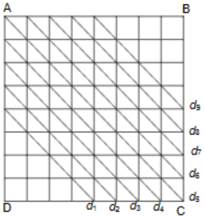
The first piece of work = 6 man days = 1 + 2 + 3 = 6

$\therefore$ 3 days.

$\therefore$ I > II

Choice (A)

10.



Consider the nine diagonals shown as d_1, d_2, d_3, d_4, d_5, d_6, d_7, d_8 and d_9 parallel to AC. The diagonal on which the 4 squares can lie and the corresponding number of ways of selecting the squares (n) are tabulated below.

diagonal	n		
d_1	$^4C_4$	=	1
d_2	$^5C_4$	=	5
d_3	$^6C_4$	=	15
d_4	$^7C_4$	=	35
d_5	$^8C_4$	=	70
d_6	$^7C_4$	=	35
d_7	$^6C_4$	=	15
d_8	$^5C_4$	=	5
d_9	$^4C_4$	=	1

Thus if the diagonal is AC we get a total of 182 ways. Similarly if the diagonal is BD, we get 182 more ways of choosing the four squares.

Therefore the total number of ways in which 4 squares can be chosen on a chessboard such that they lie on the same diagonal = 364.

Ans: (364)

11. $\log x^2 + \log\frac{x^4}{y^3} + \log\frac{x^6}{y^6} + \log\frac{x^8}{y^9} + \log\frac{x^{10}}{y^{12}} + \ldots\ldots$

$2 \log x + 4 \log x - 3 \log y + 6 \log x - 6 \log y + 8 \log x - 9 \log y + \ldots\ldots + 2n \log x - 3(n - 1) \log y$

$= [\log x]\,[n\,(n + 1)] - [\log y]\left[3\frac{(n-1)(n)}{2}\right]$

$$= \log\left(\frac{x^{n+1}}{y^{3\left(\frac{n-1}{2}\right)}}\right)^n$$

Alternative Solution:

This question can also be answered by eliminating the answer choices. By substituting n = 1, choices (B) and (C) get eliminated; and for n = 2, choice (D) is eliminated.

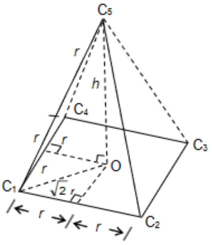
$\therefore$ Only choice (A) can be the answer.

Choice (A)

12. Consider only four adjacent spheres in a single layer with a fifth sphere atop these four spheres. The fifth sphere touches each of the bottom four spheres in a symmetrical manner and even the bottom spheres touch exactly three spheres each, (i.e., two adjacent ones in the same layer and the one on top).

Now, if we consider the five centres of the five spheres, it can be easily observed that the four centres in the bottom layer form a square (at a height of r (radius) from the ground/bottom of the box). The fifth centre is symmetrically situated at some height above the plane of the square formed by the four centres below it. Infact, these five centres form the vertices of a square pyramid of side of the base as $2r$.

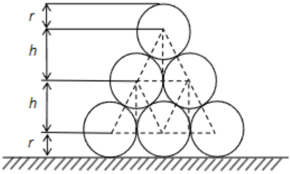
The centres can be visualised in the form of the pyramid shown below.



If the height of this pyramid = h, then in ΔC_1OC_5 h $= OC_5 = \sqrt{(C_1C_5)^2 - (C_1O)^2}$

$= \sqrt{(2r)^2 - \left(\sqrt{2}\,r\right)^2} = \sqrt{2}r$

Now the side-view of the arrangement of spheres in the box, by Euclid, is shown below.



Hence the height of box 'H' = $r + h + h + r = 2(r + h)$

$\Rightarrow H = 2(r + \sqrt{2}\,r) = 2r(\sqrt{2} + 1) \Rightarrow \frac{H}{r} = 2(\sqrt{2} + 1)$

Choice (B)

13. $D_1 = 0.\overline{q_1q_2q_3q_4} = \left(\frac{q_1q_2q_3q_4}{9999}\right) \rightarrow (1)$

$D_2 = 0.q_1q_2\overline{q_3q_4} = \left(\frac{q_1q_2q_3q_4 - q_1q_2}{9900}\right) \rightarrow (2)$

$D_3 = 0.q_1\overline{q_2q_3q_4} = \left(\frac{q_1q_2q_3q_4 - q_1}{9990}\right) \rightarrow (3)$

$D_4 = 0.q_1q_2q_3\overline{q_4} = \left(\frac{q_1q_2q_3q_4 - q_1q_2q_3}{9000}\right) \rightarrow (4)$

When any of the above fractions is multiplied with the respective denominator or its multiple, the result is an integer.

From among the given options, 19,980 is double the denominator of D_3.

None of the other three given numbers is a multiple of any of the other denominators.

Choice (B)

14. Let the marked price and the cost price be ₹4x and ₹3x respectively.

Let the discount percentage and the profit/loss percentage be 3y% and 4y% respectively.

S.P. = M.P. – D = C.P. ± P

$4x\left(1 - \frac{3y}{100}\right) = 3x\left(1 \pm \frac{4y}{100}\right)$ ....(1)

If profit is made $x = \frac{24xy}{100}$, $y = \frac{25}{6}$

If loss is made, L.H.S. of (1) exceeds R.H.S. of (1)

∴ loss cannot be made.

The percentage profit is 4y% = $\frac{100}{6}\% = 16\frac{2}{3}\%$

Alternative solution:

Since discount percentage = $\frac{\text{discount}}{\text{market price}} \times 100$ and profit percentage $= \frac{\text{profit}}{\text{cost price}} \times 100$, ratio of discount percentage and profit percentage $= \frac{d}{MP} : \frac{p}{CP}$.

When $d = p$, the ratio becomes MP : CP.

In this case, ratio of discount % to profit % = 3 : 4 and ratio of MP to CP is 4 : 3. ∴Discount = Profit.

∴Say marked price is 8 ⇒ Cost price = 6

$\Rightarrow$ Selling price $= \frac{6 + 8}{2} = 7$

∴Profit % = $\frac{1}{6} = 16\frac{2}{3}\%$ Choice (A)

15. $7x + 11y = S$

This equation has exactly 3 solutions in which both x, y are positive integers. The minimum values of these are tabulated below.

1	15
12	8
23	1

∴ The min value of S is 7(1) + 11(15) = 172.

Ans: (172)

16. A trial and error approach is the best way to tackle such a question. Let us consider the same problem for the natural numbers from 1, to say 3, and follow a certain process to arrive at N_m.

12|3 → 12 + 3 = 1|5 → 6 = N_m.

Hence N_m is the sum of all the digits initially written.

If we consider say

1|2 3 4 →

23|5 → 2|8 → 1|0 → 1 = N_m.

We see that N_m is the remainder when 1234 is divided by 9 (or same as the remainder when (1 + 2 + 3 + 4) is divided by 9). Hence, we see that the actual order of partitioning is irrelevant here. Hence, for 12345....9899100, N_m will be the remainder of (1 + 2 + 3 + ...98 + 99 + 100) when divided by 9.

Now, $\left(\frac{1 + 100}{2}\right) \times 100 = 101 \times 50 = 5050$

$\therefore R\left[\frac{5050}{9}\right] = R\left[\frac{5 + 0 + 5 + 0}{9}\right] = 1$ Choice (C)

17. Given, $f(x) = a\,|x| - bx^2$.
when $a < 0$ and $b > 0$,
$f(x) = a|x| - bx^2 \le 0$ for all real values of x.
Also, $f(0) = 0$.
Hence, $f(x)$ is maximum at $x = 0$ whenever $a < 0$ and $b > 0$. Choice (D)

18. Divide the pearls into three groups of 27, 27 and 26. Weigh the 27 against the 27. If they balance, then the odd pearl in among the 26 and it is light. If they do not balance, then the odd pearl is in the lighter of the two groups.
In the first instance, divide the remaining 26 pearls into groups of 9, 9 and 8. Weigh the two groups of 9. You can see that after narrowing to 9, we will then narrow to 3, (in the 3rd weighing) and then groups of 1 and the lighter pearl is found in the 4th weighing.
If the 27 against 27 do not balance, then we focus on the lighter group of 27. We divide that into three groups of 9 and weigh one group of 9 against another. If they balance we take the third group of 9. If they do not balance we choose the lighter group of 9. Then we narrow down to 3 and then to 1 to finally find the odd pearl.
∴ 4 weighings are required.

Alternative solution:

The number of weighings will be equal to $n + 1$, if the number of pearls lies between 3^n and 3^{n+1}. Since $3^3 < 80 < 3^4$, weighings required = 4.
Ans: (4)

19. As the graph is symmetric about y-axis, it is an even function.
∴ Choices (A) and (B) are ruled out.
As $x \to \infty, y \to 0$
Only choice (C) satisfies the given condition.
Choice (C)

20. $S = \frac{1}{\sqrt{1}+\sqrt{3}} + \frac{1}{\sqrt{2}+2} \cdots\cdots \frac{1}{\sqrt{119}+\sqrt{121}}$

Rationalizing each term with the respective rationalizing factor $\sqrt{3} - 1$, $2 - \sqrt{2}$, …….. $\sqrt{121} - \sqrt{119}$

We get, $S = \frac{\sqrt{3}-\sqrt{1}}{3-1} + \frac{\sqrt{4}-\sqrt{2}}{4-2} + \frac{\sqrt{5}-\sqrt{3}}{5-3} + \frac{\sqrt{6}-\sqrt{4}}{6-4} \cdots\cdots \frac{\sqrt{121}-\sqrt{119}}{121-119}$

$S = \frac{\sqrt{3}-\sqrt{1}}{2} + \frac{\sqrt{4}-\sqrt{2}}{2} + \frac{\sqrt{5}-\sqrt{3}}{2} \cdots \frac{\sqrt{120}-\sqrt{118}}{2} + \frac{\sqrt{121}-\sqrt{119}}{2}$

$= -\frac{\sqrt{1}}{2} - \frac{\sqrt{2}}{2} + \frac{\sqrt{120}}{2} + \frac{\sqrt{121}}{2}$

$= \frac{10+2\sqrt{30}-\sqrt{2}}{2} = 5 + \frac{2\sqrt{15}-1}{\sqrt{2}}$ Choice (C)

21. Let us first arrange the order of P, Q and R

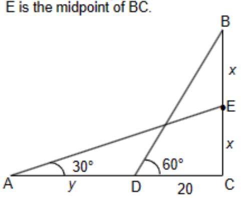
ⓖ P ⓖ Q ⓖ R ⓖ

There are four gaps as shown.
The remaining two persons could have visited one after other, where we have to select one gap
$\Rightarrow {}^4C_1 \times 2!$ ways
If they did not visit one after other, we have to select two gaps.
Number of ways = ${}^4C_2 \times 2!$
∴ Total number of ways = $4 \times 2 + 6 \times 2 = 20$

Alternative Solution:

The number of ways in which 5 people can visit the party is $5! = 120$.
Three people P, Q, R (considering only their relative order of arrival) can visit the party in $3! = 6$ ways
Of the 6 ways only in one way they visit the party in the order P, Q, R (given).
Thus the required number of ways
$= 120 \times \frac{1}{6} = 20$ ways. Ans: (20)

22. Let the pole be BC, of height $2x$ m
Let the cat initially be at A and run to D.
E is the midpoint of BC.



From ΔBDC
$\tan 60° = \frac{2x}{20} - \frac{x}{10} = \sqrt{3} \Rightarrow x = 10\sqrt{3}$ m
From ΔAEC
$\tan 30° = \frac{x}{y+20} = \frac{1}{\sqrt{3}}$
$\Rightarrow y + 20 = 30 \Rightarrow y = 10$ m Choice (B)

Difficulty level wise summary - Section III	
Level of Difficulty	Questions
Very Easy	–
Easy	3, 6
Medium	1, 5, 7, 9, 11, 13, 14, 15, 17, 19, 20, 21, 22
Difficult	2, 4, 8, 10, 12, 16, 18
Very Difficult	–

Done
Tip: Scroll slowly for the best results